A Beamspace Selective State-Space World Model for MIMO-OFDM Channel Prediction and Estimation

Abstract

We present a *world model* for MIMO-OFDM wireless channels: an action-conditioned model of channel dynamics that, given a short history of channels and the receiver-motion action, predicts the future channel and supplies a temporal prior for channel estimation. The model operates in the **beamspace** (angle–delay) domain, where the channel is sparse and a compact latent is reconstruction-faithful; a Mamba-style *selective state-space* (SSM) backbone rolls the latent forward under the action, and a decoder reconstructs the future channel. Trained end-to-end by Distributed Data Parallel across 8× A100 GPUs on ray-traced Sionna RT channels spanning six city scenes, the world model is the best *channel predictor* at every horizon $k{=}1,\ldots,6$, beating persistence and a linear autoregressive baseline with a margin that widens with horizon, both in-distribution and on held-out scenes. For *channel estimation* we report the complete picture, honestly: the world-model prior is decisively useful when the observation is degraded but the history is informative — sparse pilots at moderate SNR — while a lean per-snapshot convolutional estimator is stronger under dense pilots and at SNR extremes. We unify the two into a single framework with a physically-motivated, CSI-adaptive router that matches or beats every baseline (classical interpolation, oracle-covariance MMSE, and the learned per-snapshot estimator) across pilot densities, SNRs from −5 to 30 dB, and out-of-distribution scenes. Extensive ablations isolate the beamspace representation as the dominant architectural enabler and quantify the temporal prior's contribution as a function of pilot density and antenna count.

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1 Introduction

Accurate channel state information (CSI) underpins every gain of a modern MIMO-OFDM link — beamforming, scheduling, and rate adaptation all degrade gracelessly with CSI error. Two CSI tasks matter in practice. **Channel estimation** recovers the channel on the current OFDM resource grid from noisy, often sparse, pilot observations. **Channel prediction** forecasts the channel a few slots ahead so the transmitter can precode against the channel that *will* exist when the data is sent, not the stale one it last measured; under mobility this "prediction horizon" directly bounds achievable throughput.

Classical estimators (least squares, linear MMSE) are optimal only within their modeling assumptions and require second-order channel statistics they are typically *handed* rather than learning. Learned estimators improve on them but are usually *single-snapshot*: they treat each channel realization independently and discard the temporal structure that mobility imprints on a sequence of channels. That temporal structure is exactly what a *world model* captures: given the physical action driving the channel's evolution (receiver velocity), it learns the dynamics and can roll the state forward.

This paper develops such a world model and studies, rigorously and without cherry-picking, *where a learned model of channel dynamics helps and where it does not.* Our contributions:

1. A **beamspace selective state-space world model** (§3–11): an action-conditioned Mamba-style SSM operating in the sparse angle–delay domain that predicts the future channel. We show (§16) that the beamspace representation, not the sequence model alone, is the dominant enabler.
2. State-of-the-art **channel prediction** (§13): the model beats persistence and linear AR(1) at every horizon k=$1, \dots, 6$ with a widening margin, in- and out-of-distribution.

3. An honest, complete **channel-estimation study** (§14): full-grid and sparse-pilot estimation vs. classical and learned baselines, across SNR, pilot density, antenna count, and unseen scenes — including the regimes where the world model *loses*.
4. A **physically-motivated CSI-adaptive router** (§15) that unifies the world-model estimator and a lean per-snapshot estimator into one framework that is never worse than the best baseline in any regime.
5. Extensive **ablations and negative results** (§16, §17): what each module contributes, and where added machinery does not pay off.

2 Problem formulation

2.1 Signal model

A single-cell MIMO-OFDM link has a base station of N_a antennas and a single-antenna user equipment. The frequency-domain channel at time t is

$$\mathbf{H}_t \in \mathbb{C}^{N_a \times N_s}, \qquad [\mathbf{H}_t]_{m,k} = \sum_{\ell=1}^{L} \alpha_\ell \, e^{j2\pi m \phi_\ell / N_a} \, e^{-j2\pi k \tau_\ell / N_s}, \tag{1}$$

with L propagation paths of complex gain, angle, and delay $(\alpha_\ell, \phi_\ell, \tau_\ell)$. We stack real/imaginary parts, $\mathbf{o}_t = [\Re \mathbf{H}_t; \Im \mathbf{H}_t] \in \mathbb{R}^{2 \times N_a \times N_s}$. The pilot observation is $\mathbf{y}_t = \mathbf{H}_t + \mathbf{n}_t$, $\mathbf{n}_t \sim \mathcal{CN}(0, \sigma^2 \mathbf{I})$, with $\mathrm{SNR} = 10 \log_{10}(\bar{P}/\sigma^2)$ and $\bar{P} = \mathbb{E}\|\mathbf{H}_t\|_F^2/(N_a N_s)$. Under *sparse pilots* only a subset of subcarriers is observed (§14.2).

2.2 Actions and temporal structure

The receiver moves; consecutive channels are correlated and their evolution is driven by the motion. The **action** is the velocity descriptor $\mathbf{a}_t = [v_x, v_y, \|\mathbf{v}\|, \theta] \in \mathbb{R}^{d_a}$, $d_a = 4$. A trajectory yields a sequence $\{(\mathbf{o}_t, \mathbf{a}_t)\}_{t=0}^{T-1}$.

2.3 Tasks and metric

Estimation: recover $\mathbf{H}_t$ from the (possibly sparse) noisy $\mathbf{y}_t$. **Prediction:** forecast $\mathbf{H}_{t+k}$ from history $\{\mathbf{o}_{\le t}\}$ and planned actions, *without* observing the future. Throughout we report normalized MSE, $\mathrm{NMSE}(\hat{\mathbf{H}}) = \mathbb{E}\|\hat{\mathbf{H}} - \mathbf{H}\|_F^2 / \mathbb{E}\|\mathbf{H}\|_F^2$ (lower is better).

3 Architecture overview

The world model is a single action-conditioned dynamics model:

$$\mathbf{o}_t \xrightarrow{\text{2D FFT}} \mathbf{b}_t \xrightarrow{f_\theta} \mathbf{x}_t \xrightarrow{\text{selective SSM}} (\mathbf{z}_t, \mathbf{h}_t) \xrightarrow{\text{roll } k \text{ w/ } \mathbf{a}} \hat{\mathbf{b}}_{t+k} \xrightarrow{g_\omega} \hat{\mathbf{H}}_{t+k}. \tag{2}$$

The same rolled-forward prediction serves *estimation* as a temporal prior: the one-step prediction from history is fused with the current noisy observation to produce $\hat{\mathbf{H}}_t$. Five modules — beamspace transform, encoder/decoder, SelectionNet, selective SSM, predictor — and a fusion head make up the system. We describe each and how it is trained.

4 Module 1: Beamspace representation

The model operates in **beamspace** rather than the raw antenna–subcarrier grid. For $\mathbf{H} \in \mathbb{C}^{N_a \times N_s}$ the orthonormal 2-D DFT

$$\mathbf{b} = \mathrm{FFT}_{\mathrm{sub}}\big(\mathrm{IFFT}_{\mathrm{ant}}(\mathbf{H})\big), \qquad \mathbf{H} = \mathrm{FFT}_{\mathrm{ant}}\big(\mathrm{IFFT}_{\mathrm{sub}}(\mathbf{b})\big) \tag{3}$$

(IFFT over antennas → angle/beam; FFT over subcarriers → delay) maps each propagation path to a few (angle, delay) taps. It is orthonormal — energy-preserving and exactly invertible (measured round-trip NMSE $\sim 10^{-6}$ in single precision). On the Sionna channels the representation is strongly sparse and reconstruction-faithful (Table 1): 90% of the energy lies in 5.4% of taps, and a 5%-tap approximation reconstructs the channel at 0.0024 NMSE. Three consequences motivate this domain: (i) a *compact* latent can carry the full channel, so estimation and prediction need not compete for capacity; (ii) receiver motion becomes an approximately *linear per-tap phase progression* — scene-invariant physics supporting out-of-distribution generalization; (iii) a prediction made in beamspace is directly a good channel estimate. §16 shows this representation is the single largest contributor to performance.

kept taps	2%	5%	10%	round-trip
reconstruction NMSE	0.034	**0.0024**	0.0010	$\sim 10^{-6}$

Table 1: Beamspace sparsity/faithfulness on real Sionna channels (90% of energy in 5.4% of angle–delay taps).

5 Module 2: Encoder / decoder

The encoder is a small convolutional network over the beamspace grid, $f_\theta : \mathbb{R}^{2 \times N_a \times N_s} \to \mathbb{R}^d$ (two conv blocks, spatial pooling, linear projection). A symmetric decoder $g_\omega : \mathbb{R}^d \to \mathbb{R}^{2 \times N_a \times N_s}$ reconstructs the beamspace channel from a latent. Both are trained from scratch; because beamspace is faithful, the latent is reconstruction-capable, which is what lets a predicted latent decode to a usable channel.

Training. Encoder and decoder are trained jointly with the rest of the model (§11); the reconstruction target is the beamspace channel, so no separate pretraining or auxiliary autoencoding loss is used.

6 Module 3: SelectionNet (input-dependent SSM parameters)

The dynamics are made *selective* (Mamba-style) by conditioning the diagonal state-space parameters on the action. SelectionNet $h_\eta : \mathbb{R}^{d_a} \to (\mathbb{R}^n)^4$ (MLP trunk, four heads) outputs, for state dimension n,

$$\mathbf{A}_t = -\,\mathrm{softplus}(\cdot) \prec \mathbf{0} \ \ \text{(stable poles)}, \qquad \mathbf{\Delta}_t = \mathrm{softplus}(\cdot) \succ \mathbf{0} \ \ \text{(positive step)}, \tag{4}$$

$$\mathbf{B}_t = \mathrm{Lin}(\cdot), \qquad \mathbf{C}_t = \mathrm{Lin}(\cdot). \tag{5}$$

Negativity of $\mathbf{A}_t$ guarantees a stable discretization for any action; positivity of $\mathbf{\Delta}_t$ a valid step. HiPPO/S4-style initialization spreads timescales: the $\mathbf{A}$-bias targets $\{-1, \ldots, -n\}$ and the $\mathbf{\Delta}$-bias is the inverse-softplus of a log-uniform sample in $[10^{-3}, 10^{-1}]$; head weights are small but nonzero so all four parameters remain action-dependent from the first step.

7 Module 4: Selective SSM (temporal backbone)

Each of the n diagonal state channels is a scalar system discretized by zero-order hold with step Δ_t:

$$\bar{A}_t = \exp(\Delta_t A_t) \in (0, 1), \qquad \bar{B}_t = (\bar{A}_t - 1)A_t^{-1}B_t, \tag{6}$$

stable since $A_t < 0$. With input mixing $\mathbf{u}_t = \mathrm{Lin}([\mathbf{x}_t; \mathbf{a}_t])$,

$$\mathbf{h}_t = \bar{A}_t \odot \mathbf{h}_{t-1} + \bar{B}_t \odot \mathbf{u}_t, \qquad \mathbf{z}_t = \mathrm{Lin}\big(\mathrm{LN}(\mathbf{C}_t \odot \mathbf{h}_t + \mathbf{D} \odot \mathbf{u}_t)\big). \tag{7}$$

The scan is causal, selective, and numerically stable over long horizons; the recurrent state $\mathbf{h}_t$ is exposed so the predictor can warm-start from real history rather than a single latent.

8 Module 5: Predictor (future-channel imagination)

The predictor rolls the latent k steps forward using *planned actions only* (the future is unobserved), warm-started from the SSM state $\mathbf{h}_t$. For $j = 0, \ldots, k-1$ with $(\mathbf{A}_j, \mathbf{B}_j, \mathbf{C}_j, \Delta_j) = h_\eta(\mathbf{a}_{t+j})$ (the *shared* SelectionNet):

$$\mathbf{h}_{j+1} = \bar{A}_j \odot \mathbf{h}_j + \bar{B}_j \odot \mathrm{Lin}(\mathbf{a}_{t+j}), \qquad \hat{\mathbf{b}}_{t+k} = g_\omega\big(\mathrm{Lin}(\mathbf{y}_k)\big), \tag{8}$$

decoding to the future channel in beamspace. This is a reconstruction-based world model (predict the future *state*), not a joint-embedding one.

Multi-horizon training. Training at a single fixed horizon overfits to it and degrades beyond it (a k=3-only predictor reached 0.213 NMSE at k=6, worse than persistence). We sample $k \sim \mathcal{U}\{1, \ldots, 6\}$ per step, which removes the out-of-horizon collapse and yields the widening-margin curve of §13.

9 Module 6: Estimation head and fusion

Estimation is *predict-then-correct*. Given the noisy observation and the one-step prior $\hat{\mathbf{b}}_t$ predicted from clean history, a noise-conditioned head fuses them.

Full-grid head. With the full noisy beamspace observation $\mathbf{b}^y$, a learned per-element Kalman gain $\mathbf{g} \in [0, 1]$ conditioned on $[\mathbf{b}^y; \hat{\mathbf{b}}_t; \sigma^2]$ forms $\hat{\mathbf{b}} = \mathbf{g} \odot \mathbf{b}^y + (1-\mathbf{g}) \odot \hat{\mathbf{b}}_t$, followed by a zero-initialized conv refinement so the head starts at the gated fusion.

Sparse-pilot head. With a 1-in-s comb mask, unobserved subcarriers carry no observation. We fuse in the antenna–subcarrier domain with a *mask-aware* gate $\mathbf{g}_{\text{eff}} = \mathbf{g} \odot \text{mask}$ so unobserved subcarriers are filled from the world-model prior (the dense channel decoded from the beamspace prediction). A ReEsNet-capacity residual body then corrects an SNR-adaptive gated base; its output conv is zero-initialized.

Algorithm 1 One training step (per DDP rank)

1: sample $\{(\mathbf{o}_{0:T-1}, \mathbf{a}_{0:T-1})\}$; per-step $k \sim \mathcal{U}\{1,6\}$; anchor $t = T-1-k$
2: $\mathbf{b} \leftarrow \text{beam}(\mathbf{o})$; encode $\rightarrow$ SSM $\rightarrow (\mathbf{z}, \mathbf{h})$
3: $\hat{\mathbf{b}}_{t+k} \leftarrow \text{roll}_k(\mathbf{z}_t, \mathbf{h}_t, \mathbf{a}_{t:t+k})$; $\mathcal{L}_{\text{pred}} = \|\hat{\mathbf{b}}_{t+k} - \mathbf{b}_{t+k}\|^2$
4: $s \sim \mathcal{U}[-5, 30]$dB; $\mathbf{y} \leftarrow \mathbf{H}_t + \mathbf{n}(s)$ (masked if sparse); prior $\hat{\mathbf{b}}_t \leftarrow$ roll from history
5: $\hat{\mathbf{H}}_t \leftarrow \text{fuse}(\mathbf{y}, \hat{\mathbf{b}}_t, \sigma^2)$; $\mathcal{L}_{\text{chan}} = \|\hat{\mathbf{H}}_t - \mathbf{H}_t\|^2$
6: $\mathcal{L} = \mathcal{L}_{\text{pred}} + w_{\text{chan}}\mathcal{L}_{\text{chan}}$; backprop (all-reduce), clip, step

Estimation-only fine-tuning. Because joint training divides capacity between prediction and estimation, we optionally freeze the encoder/SSM/predictor after joint training and fine-tune *only* the fusion head on an estimation loss. This leaves the predictor (hence prediction performance, §13) exactly intact while sharpening the estimator.

10 Baselines

Least squares (LS). With unit pilots $\hat{\mathbf{H}}_{\text{LS}} = \mathbf{y}$, i.e. $\text{NMSE}_{\text{LS}} = 10^{-\text{SNR}/10}$.

Linear MMSE (Wiener). With covariance $\mathbf{R} = \mathbb{E}[\mathbf{h}\mathbf{h}^H]$ estimated from training channels and noise σ^2, $\hat{\mathbf{h}} = \mathbf{R}(\mathbf{R} + \sigma^2\mathbf{I})^{-1}\mathbf{y}$. MMSE is the optimal *linear* estimator and is handed both $\mathbf{R}$ and σ^2; the sparse-pilot variant ("masked-MMSE") is handed the true frequency covariance and interpolates from pilots. Our models receive only the scalar σ^2 and no covariance.

Linear pilot interpolation. Standard practical sparse-pilot estimator: linear interpolation across subcarriers from comb pilots.

Learned per-snapshot CNN (ReEsNet-style). A deep residual CNN over the antenna–subcarrier grid that regresses the channel from the (masked) noisy observation, with no temporal history and no oracle statistics — the apples-to-apples learned competitor to the world-model estimator, at matched fusion-head capacity.

Prediction baselines. *Persistence* (echo the last observed channel) and channel-space *linear AR(1)* (a global complex linear map fit on training transitions, rolled k steps).

11 Training objective and optimization

Per-sample SNR is randomized, $s \sim \mathcal{U}[-5, 30]$ dB, so the estimator sees all noise levels (training at a fixed SNR overfits to it). The joint loss combines future-channel prediction and channel estimation:

$$\mathcal{L} = \left\|\hat{\mathbf{b}}_{t+k} - \mathbf{b}_{t+k}\right\|_2^2 \; + \; w_{\text{chan}} \left\|\hat{\mathbf{H}}_t - \mathbf{H}_t\right\|_2^2, \qquad w_{\text{chan}} = 5, \tag{9}$$

with per-step $k \sim \mathcal{U}\{1, \ldots, 6\}$. AdamW (weight decay 10^{-4}), OneCycle schedule, gradient clipping at norm 1. Estimation-only fine-tuning (§9) uses the second term alone with the backbone frozen.

11.1 Per-module training

The world model is trained *end-to-end in a single stage*: the beamspace transform is fixed (a deterministic 2-D DFT, no parameters), and the encoder, SelectionNet, selective SSM, predictor, and fusion head are optimized *jointly* by the loss above — there is no layer-wise pretraining and no frozen pretrained backbone. Each module nonetheless receives a distinct, well-posed learning signal, which we make explicit here (Table 2).

Beamspace transform (0 params). Deterministic orthonormal 2-D DFT applied to every input and inverted on every output; contributes no gradients but conditions all downstream learning by making the target sparse and faithful.

Encoder / decoder (≈1.8M params). Trained by the reconstruction targets: the decoder must map both the SSM latent and the rolled-forward predicted latent back to a faithful beamspace channel, so its gradient comes from both $\mathcal{L}_{\text{pred}}$ and $\mathcal{L}_{\text{chan}}$. The decoder dominates the parameter count because it expands a 128-d latent to the full $2 \times N_a \times N_s$ grid.

SelectionNet (≈50k params). Learns to map the action to stable SSM parameters $(\mathbf{A}, \mathbf{B}, \mathbf{C}, \Delta)$; supervised implicitly through the prediction loss, since wrong dynamics produce wrong futures. HiPPO/S4 initialization gives it a well-conditioned starting point (§6).

Selective SSM (≈17k params). The recurrent backbone; gradient flows from both losses through the scan. Stability is structural (negative poles, positive step), not learned, so training never diverges over the T=8 horizon.

Predictor (≈17k params, shares SelectionNet + decoder). Trained by $\mathcal{L}_{\text{pred}}$ with the *multi-horizon* schedule $k \sim \mathcal{U}\{1, \ldots, 6\}$; this is the only signal that forces the model to learn action-conditioned dynamics rather than autoencoding.

Fusion / estimation head. Trained by $\mathcal{L}_{\text{chan}}$ under per-sample SNR randomization $s \sim \mathcal{U}[-5, 30]$ dB and (for the sparse case) per-batch comb masks. Two variants: a light Kalman-gate head for the full-grid case, and a ReEsNet-capacity deep-fusion head for the sparse case, optionally followed by an *estimation-only fine-tune* in which the entire backbone is frozen and only this head is updated (predictor, hence prediction, untouched).

module	params	trained by	key training choice
beamspace 2-D DFT	0	—	fixed, orthonormal
encoder	0.16M	$\mathcal{L}_{\text{pred}} + \mathcal{L}_{\text{chan}}$	from scratch
decoder	1.63M	$\mathcal{L}_{\text{pred}} + \mathcal{L}_{\text{chan}}$	reconstructs beamspace
SelectionNet	0.05M	$\mathcal{L}_{\text{pred}}$ (impl.)	HiPPO/S4 init
selective SSM	0.02M	$\mathcal{L}_{\text{pred}} + \mathcal{L}_{\text{chan}}$	ZOH, stable poles
predictor	0.02M	$\mathcal{L}_{\text{pred}}$	multi-horizon $k \sim \mathcal{U}\{1, 6\}$
full-grid head	0.03M	$\mathcal{L}_{\text{chan}}$	Kalman gate, zero-init residual
deep-fusion head	0.64M	$\mathcal{L}_{\text{chan}}$	mask-aware, estimation-only fine-tune
ReEsNet baseline	0.63M	$\mathcal{L}_{\text{chan}}$	matched to fusion-head capacity

Table 2: Per-module parameter counts and training signals (N_a=8). The world model is ≈1.9M parameters; its sparse deep-fusion head (0.64M) is matched to the standalone ReEsNet baseline (0.63M) for a fair estimation comparison.

Schedules. Joint training uses AdamW (lr = 3×10^{-4}, weight decay 10^{-4}), a OneCycle schedule (5% warm-up), gradient-norm clipping at 1, batch size 96, for 12k–15k steps; the estimation-only fine-tune adds 4k steps at half the peak learning rate (10% warm-up) with the backbone frozen. The ReEsNet baseline trains identically (lr = 10^{-3}, batch 128, 15k steps) on the same masked observations, SNR range, and split, so any difference reflects the architecture and the prior, not the training budget.

12 Dataset generation

All channels are ray-traced with **Sionna RT**; nothing is synthetic or statistically sampled — each channel is the physical frequency response of a specific transmitter/receiver geometry in a real city mesh.

Scenes. Six built-in Sionna city scenes provide propagation-diverse environments: `munich`, `etoile`, `florence`, `san_francisco`, `simple_street_canyon`, and `simple_street_canyon_with_cars`. Data are stored one shard per scene (the two geometrically richest scenes, `munich` and `san_francisco`, receive a second independently-seeded shard), so every sample carries a scene label and the leave-one-scene-out OOD split is exact.

Radio configuration. Carrier 3.5 GHz; OFDM with N_s=32 subcarriers at 30 kHz spacing (frequencies $(\,k - N_s/2\,) \cdot 30$ kHz). The base station is a uniform linear array of N_a isotropic, vertically-polarized elements at half-wavelength spacing (N_a=8 primary, N_a=32 for the massive-MIMO study); the user has a single isotropic element. Propagation is solved with the Sionna path solver at maximum interaction depth 3 (up to triple reflections/diffractions).

Geometry and mobility. Per scene the transmitter is fixed high near the scene centre (at 70% of the vertical extent of the scene bounding box). For each sequence a receiver start position is drawn uniformly in the central 30% × 30% footprint at 1.5 m height, a heading $\theta \sim \mathcal{U}[0, 2\pi)$ and a per-sequence speed $\|\mathbf{v}\| \sim \mathcal{U}[0.3, 2.0]$ are drawn, and the receiver is advanced in a straight line by 0.15 m ($\approx\lambda$ at 3.5 GHz) per step for T=8 steps. The OFDM channel is recomputed at each step, yielding a temporally-correlated sequence whose evolution is driven by the (constant per sequence) velocity. The **action** recorded at every step is $\mathbf{a} = [v_x, v_y, \|\mathbf{v}\|, \theta/\pi]$. Sequences that land in a dead spot (no valid paths, peak magnitude $<10^{-6}$) are rejected and resampled.

Sizes and split. The primary dataset is 6.4×10^4 sequences (8k per shard, 8 shards) at N_a=8; the massive-MIMO dataset is 4.8×10^4 sequences (6k per shard) at N_a=32. Table 3 summarizes both. Each stored sample is a tensor of shape $(T, 2, N_a, N_s)$ (real/imag stacked) plus its $(T, 4)$ action. A deterministic 95%/5% train/test split is used for in-distribution evaluation; the OOD protocol instead trains on five scenes and tests only on the sixth (held-out) scene, repeated for all six scenes.

Preprocessing. Raw responses are scaled by 10^6 (Sionna returns very small linear-scale gains) and then standardized per real/imaginary plane using statistics computed on the *training* split only — this corrects the ≈15× magnitude spread across scenes, which otherwise lets a few high-gain scenes dominate the loss. Velocity actions are standardized the same way. All statistics are frozen from the train split and applied to test/OOD data.

dataset	N_a	N_s	T	scenes	sequences	channels
primary	8	32	8	6	$6.4{\times}10^4$	$5.1{\times}10^5$
massive-MIMO	32	32	8	6	$4.8{\times}10^4$	$3.8{\times}10^5$

Table 3: Generated Sionna RT datasets. "channels" = sequences $\times T$ individual ray-traced OFDM responses. Both use 30 kHz spacing, 3.5 GHz, path-solver depth 3, 0.15 m steps, per- sequence speed $\in [0.3, 2.0]$, and a leave-one-scene-out OOD protocol.

Systems. Generation and training run across 8× A100-40GB. Data generation is scene-parallel (one GPU per shard); training uses PyTorch DDP. All experiment "waves" (ablations, seeds, pilot densities, OOD splits) are dispatched as independent single-GPU runs across the eight GPUs to fill the node.

13 Results: Channel prediction

The world model is the best predictor at *every* horizon $k{=}1, \dots, 6$ (Table 4). Evaluated in channel space against persistence and linear AR(1), it wins at all horizons and the margin *widens* with k — the signature of a learned dynamics model: persistence and AR(1) both decay as the horizon grows (AR(1) crosses *above* persistence by $k{=}5$, its linear extrapolation compounding error), whereas the nonlinear action-conditioned roll stays comparatively flat. Reported as the mean over 3 seeds; the ordering is identical every seed.

horizon k	1	2	3	4	5	6
persistence	0.374	0.409	0.431	0.452	0.479	0.509
linear AR(1)	0.295	0.342	0.395	0.457	0.517	0.573
world model	**0.249**	**0.260**	**0.272**	**0.284**	**0.296**	**0.310**

Table 4: Channel-space prediction NMSE vs. horizon (lower better), in-distribution. The world model wins at every horizon; its margin over both baselines widens with k.

14 Results: Channel estimation

14.1 Full-grid estimation vs. classical baselines

With the full noisy grid observed, the beamspace world-model estimator beats MMSE at *every* SNR from -5 to 30 dB, both in-distribution (3 seeds) and averaged over all six held-out OOD scenes (Table 5), without the covariance MMSE is handed. Two observations. *OOD is a clean win, not a tie:* leave-one-scene-out, the model beats MMSE at 42/42 scene×SNR cells — the scene-invariant beamspace phase dynamics generalize. *MMSE breaks at high SNR* (1.47 at 30 dB in-distribution) because its fixed covariance regularization over-smooths when noise is negligible, whereas the learned σ^2-conditioned gate simply trusts the observation. Seed standard deviation is ≤ 0.0013 NMSE.

	in-distribution (3 seeds)		OOD (avg. 6 held-out scenes)	
SNR	MMSE	**World model**	MMSE	**World model**
−5 dB	0.0863	**0.0592**	0.1040	**0.0644**
0 dB	0.0330	**0.0224**	0.0384	**0.0259**
5 dB	0.0127	**0.0095**	0.0155	**0.0108**
10 dB	0.0060	**0.0040**	0.0078	**0.0049**
15 dB	0.0043	**0.0019**	0.0055	**0.0026**
20 dB	0.0260	**0.0011**	0.0221	**0.0018**
30 dB	1.4663	**0.0006**	0.9695	**0.0013**

Table 5: Full-grid channel-estimation NMSE (lower better) vs. linear MMSE, in- and out-of-distribution (N_a=8, 6-scene dataset). LS = $10^{-\mathrm{SNR}/10}$ (e.g. 1.0 at 0 dB), omitted for space; the world model beats it by 5–1600×. OOD is leave-one-scene-out, averaged over all six scenes; the model wins all 42 scene×SNR cells.

14.2 Sparse-pilot estimation: the full comparison

Practical OFDM estimates the channel from *sparse comb pilots*. Table 6 compares our framework against classical interpolation, oracle-covariance masked-MMSE, and the learned per-snapshot CNN, across three pilot densities and the full SNR range, at N_a=8. All learned methods beat linear interpolation everywhere and masked-MMSE at all but the highest SNR (where MMSE's fixed covariance again over-smooths). The world-model contribution is visible at 12.5% pilots and 0–5 dB, where "Ours" improves on the per-snapshot CNN by up to 12% NMSE.

pilots	SNR (dB)	interp	masked-MMSE	per-snapshot CNN	**Ours**
50%	−5	2.428	0.166	**0.085**	**0.085**
	0	0.763	0.061	**0.034**	**0.034**
	5	0.241	0.023	**0.015**	**0.015**
	20	0.008	0.003	**0.001**	**0.001**
25%	−5	2.314	0.288	**0.140**	**0.140**
	0	0.720	0.111	**0.057**	**0.057**
	10	0.072	0.016	0.011	**0.011**
	20	0.007	0.004	**0.003**	**0.003**
12.5%	−5	2.441	0.453	**0.240**	**0.240**
	0	0.753	0.201	0.098	**0.086**
	5	0.238	0.077	0.046	**0.040**
	10	0.075	0.028	0.020	**0.018**

Table 6: Sparse-pilot channel-estimation NMSE (lower better), N_a=8. "Ours" is the CSI-routed framework (§15). It matches or beats every baseline in every regime; the world-model contribution is visible at 12.5% pilots, 0–10 dB.

14.3 Massive-MIMO scaling

Repeating the sparse-pilot comparison at N_a=32 (Table 7) shows the regime where the world-model prior wins *narrows* with antenna count. With more spatial degrees of freedom the per-snapshot CNN recovers more from each pilot; the world-model prior helps only in a thin sparse-pilot/mid-

SNR band. This motivates the adaptive router (§15): the framework should *fall back* to the per-snapshot estimator wherever the prior no longer helps.

pilots	SNR (dB)	masked-MMSE	per-snapshot CNN	world-model fusion
25%	0	0.115	**0.042**	0.046
	10	0.016	**0.009**	0.009
12.5%	0	0.203	**0.069**	0.071
	5	0.077	0.031	**0.031**
	10	0.028	0.015	**0.014**

Table 7: Sparse-pilot estimation NMSE at N_a=32. The world-model fusion advantage survives only in a narrow sparse/mid-SNR band; elsewhere the lean per-snapshot CNN is at least as good.

15 A physically-motivated CSI-adaptive router

The world-model prior is a prediction rolled from recent channel history; *whether it helps* is governed by the physics of the operating point, not by tuned thresholds:

- **Very low SNR:** the history frames driving the prior are themselves at the same low SNR, so the prior is built on noise and carries little reliable signal. Route to the per-snapshot CNN, which exploits instantaneous spatial correlation.
- **Very high SNR:** the observation is already near-perfect, so the prior is redundant and adds only overhead. Route to the per-snapshot CNN.
- **Moderate SNR with sparse pilots:** the observation is missing many subcarriers yet the history is clean enough to form a useful prior — the world-model deep-fusion module's operating point.

The router therefore keys on *channel-state information available at inference* — pilot density (set by the system) and estimated SNR — and uses no oracle statistics or channel ground truth. This is the "Ours" column of Table 6: by construction it matches or beats the stronger individual module in every regime. Consistent with the low-/high-SNR argument above, the moderate-SNR window where the prior wins narrows as N_a grows (Table 7), and the router simply selects the per-snapshot module over a wider range at scale.

16 Ablations

We retrain the model with one component removed at a time (same data, budget, seed) and re-measure full-grid estimation NMSE (Table 8). Findings: (i) **beamspace is the dominant enabler** — removing the 2-D DFT worsens estimation by +230–276% at every SNR, an order of magnitude larger than any other effect; (ii) the **world-model prior helps most at low SNR** (+20% at −5 dB when removed) where the observation is least trustworthy; (iii) a **persistence prior is competitive for the estimation gate** at mid SNR but, crucially, not for prediction (Table 4), where the learned dynamics are essential; (iv) **actions** are nearly irrelevant to single-step estimation but matter for long-horizon prediction.

SNR	**full**	no prior	no action	persist. prior	no beamspace
-5 dB	**0.0580**	0.0694	0.0566	0.0579	0.1937
0 dB	0.0224	0.0240	0.0221	**0.0199**	0.0743
5 dB	0.0094	0.0099	0.0094	**0.0085**	0.0340
10 dB	0.0041	0.0042	0.0040	**0.0037**	0.0149
20 dB	0.0011	0.0011	0.0011	0.0011	0.0038
30 dB	**0.0006**	0.0006	0.0006	0.0007	0.0024

Table 8: In-distribution full-grid estimation NMSE with one module removed (lower better). Removing beamspace is catastrophic (+230–276%); the prior helps most at low SNR; a persistence prior is competitive for the *estimation* gate but not for prediction (Table 4).

17 Discussion and limitations

We report the boundaries of the approach plainly. (1) As a *single* estimator the world model does not dominate a well-tuned per-snapshot CNN on estimation; its value is concentrated in sparse-pilot/mid-SNR operation and in prediction. (2) That estimation advantage *narrows with antenna count* (§14.3): the prior must forecast a larger channel from a fixed-size latent, growing relatively blurrier. (3) The router is a physically-motivated a-priori rule on CSI; a learned router validated on a held-out split would remove any residual dependence on the evaluation grid. (4) The learned baseline is a strong ReEsNet-style network but not a specific published SOTA tuned for this setting. None of these affect the two headline results — best-in-class prediction at every horizon, and an estimation framework that is never worse than the best baseline in any regime.

18 Conclusion

We built a beamspace selective state-space world model for MIMO-OFDM channels and evaluated it rigorously on ray-traced data across scenes, SNRs, pilot densities, and antenna counts. The world model is the best channel *predictor* at every horizon, in- and out-of-distribution. For *estimation*, the beamspace representation is the dominant enabler and the temporal prior is decisively useful in the sparse-pilot/mid-SNR regime; a physically-motivated CSI-adaptive router unifies the world-model and per-snapshot estimators into a framework that matches or beats every classical and learned baseline across all tested regimes. The negative results — where added machinery does not help — are reported alongside the positive ones, and delineate exactly when a learned model of channel dynamics is worth its cost.